

Attendance Monitoring System

Vue 3 Web Application — Technical Documentation Report

Developer: Juner Punzal
Course/Section: BSCS 3A
Technologies: Vue 3, Vite, Tailwind CSS, Local Storage

1. System Overview

The Attendance Monitoring System is a single-page web application that helps an administrator register students, record daily attendance, monitor current attendance statistics, and review individual attendance performance. The application is responsive, uses a consistent dashboard layout, and preserves data in the browser using local storage.

2. Architecture and Vue Implementation

The system is organized into reusable Vue components. `App.vue` controls page navigation, settings, dashboard totals, and reports. Feature-specific interface components such as `Students.vue` and `Attendance.vue` communicate through props and emitted events. Shared business logic is contained in composables, keeping the interface separate from data operations.

Area	Implementation
Reactivity	Vue <code>ref</code> and <code>computed</code> state automatically update lists, totals, searches, filters, and reports.
Component communication	Parent-to-child props and child-to-parent events support navigation, forms, confirmation dialogs, and notifications.
Persistence	Students, attendance records, system details, and dark-mode preference are saved in browser local storage.
Navigation	Sidebar, dashboard quick actions, header actions, and hash URLs such as <code>/#attendance</code> provide predictable navigation.

3. CRUD Features

Operation	Student Management	Attendance Management
Create	Adds a student ID, name, and course with required-field validation.	Records a student, date, status, and optional remarks.
Read	Displays a searchable student table.	Displays searchable and status-filtered records with student course information.
Update	Edits student information; an edited student ID is propagated to related attendance records.	Edits the student, date, status, or remarks of an existing record.
Delete	Removes a selected student after confirmation.	Uses a confirmation dialog before deleting a record.

Attendance validation prevents a duplicate record for the same student on the same date. Supported statuses are Present, Late, Absent, and Excused.

4. Interface Screenshots

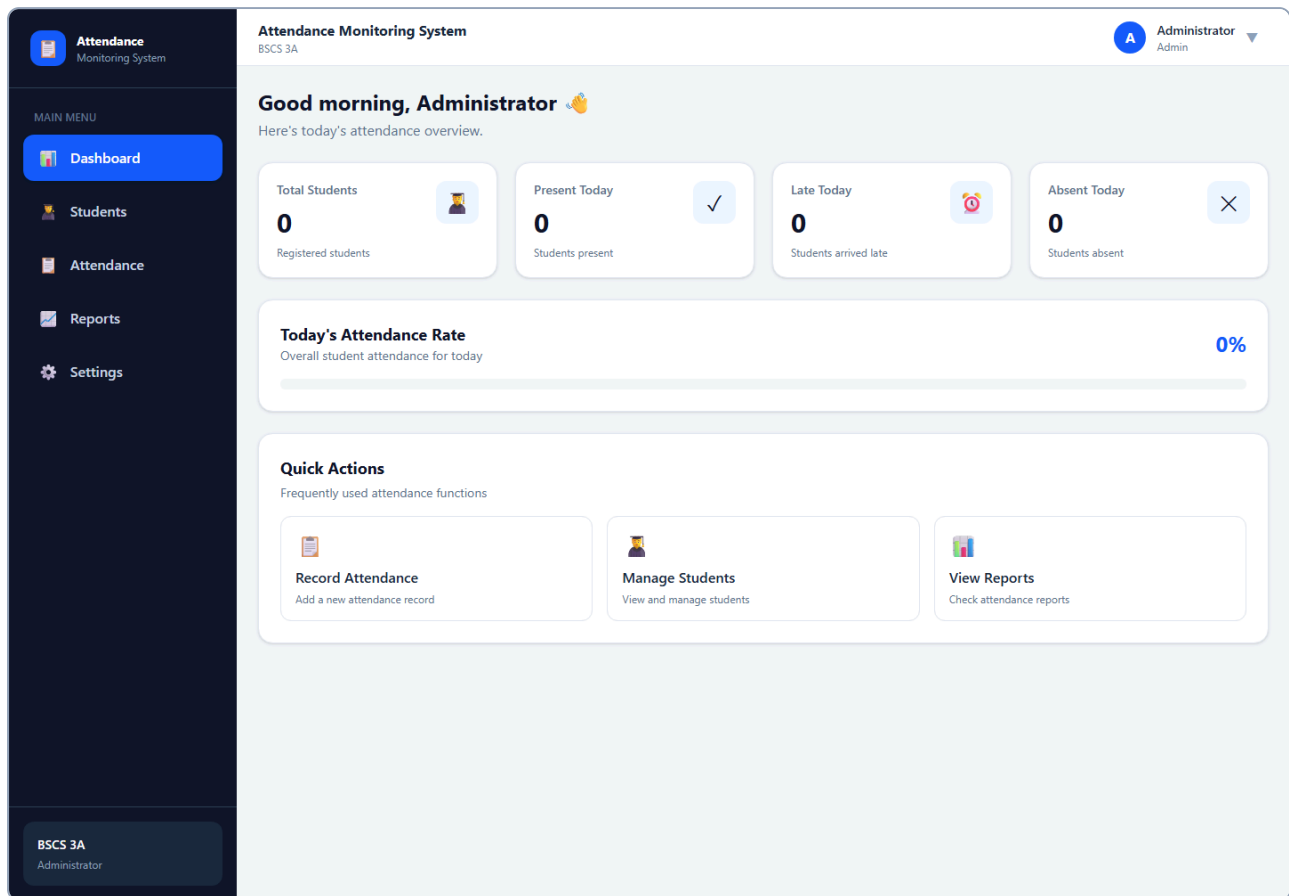


Figure 1. Dashboard showing daily metrics, attendance-rate progress, sidebar navigation, and quick actions.

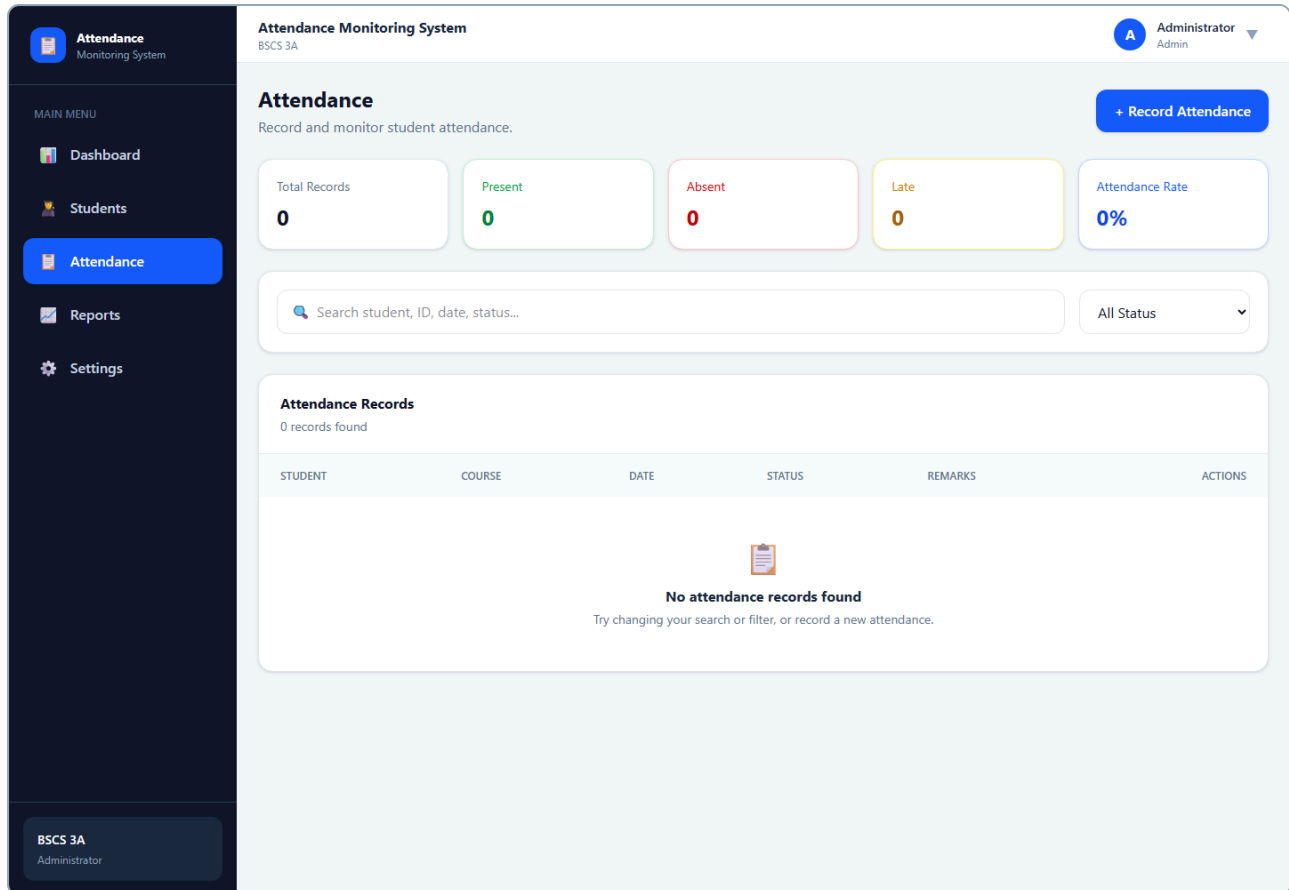


Figure 2. Attendance page with statistics, searchable/filterable record list, and record-management actions.

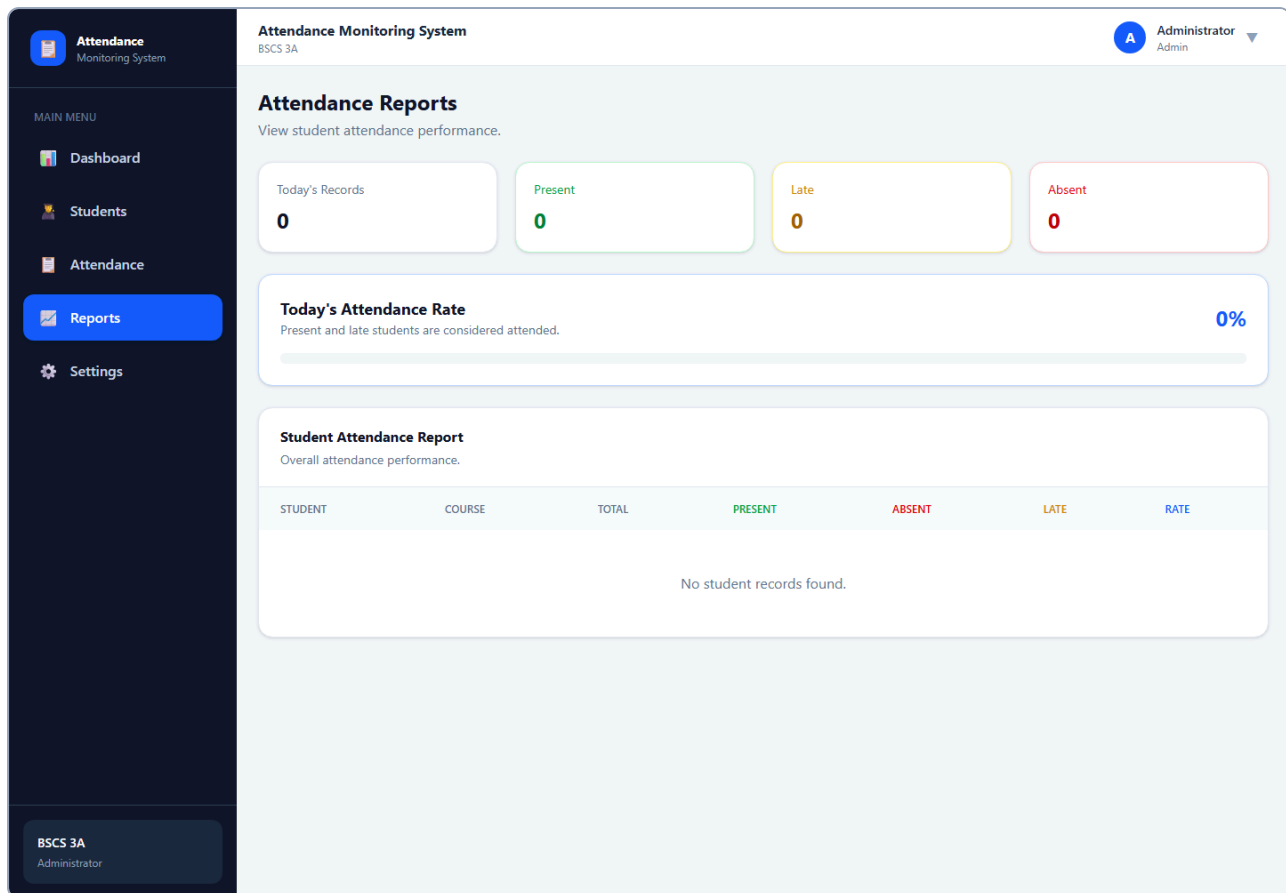


Figure 3. Reports page summarizing daily attendance and each student's overall performance.

5. Running and Building the Project

Install dependencies with `npm install`. Start development mode with `npm run dev`. Create an optimized production build with `npm run build`. The production build completed successfully during verification.

6. Version Control and Submission

The project is initialized as a Git repository with a descriptive initial commit. It is ready to be connected to a public GitHub repository. The submission package excludes `node_modules`, generated build output, and Git metadata so it is clean and easy to submit.